2024

COMPUTER SCIENCE AND ENGINEERING

Paper : CSE-903

[Wireless and Mobile Computing (Elective-II)]

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1 & 2,** and ***any four*** questions from the rest.

1. Answer ***any five*** questions : 2×5

(a) What is the structure of a cell in a cellular system? Justify the use of such a structure.

(b) List the factors that affect the performance of TCP in a wireless environment.

(c) What are the main features of M-TCP? How does it enhance TCP performance in mobile environments?

(d) Is a table-driven routing protocol suitable for high-mobility environments? Why, or why not?

(e) Why is the SPIN considered a push-based protocol?

(f) State the specific advantage of using MARCH protocol compared to existing MAC protocols for Adhoc Networks.

(g) State the role of the Network Allocation Vector in MACA protocol.

2. Answer ***any five*** questions : 4×5

(a) Describe the concept of tunneling in Mobile IP. How is encapsulation achieved during the tunneling process?

(b) Consider five wireless stations A, B, C, D, and E. Station A can communicate with all other stations. B can communicate with A, C, and E. C can communicate with A, B, and D. D can communicate with A, C, and E. E can communicate with A, D, and B. Find the set of hidden nodes and exposed nodes

(i) When A is communicating to B.

(ii) When B is communicating to A.

(iii) When B is communicating to C.

(c) Discuss the GSM Hyper-frame structure in detail.

(d) Derive a 2-Ray path loss model for the transmitter and receiver with heights of h_t meters and h_r meters, respectively, separated by a distance of d meters.

Please Turn Over

(e) "Link State routing may not be a suitable approach for a large network."— Comment on the statement and justify.

(f) Explain the contribution of MACA-W over MACA with the necessary illustration.

(g) State the purpose of the Back-off algorithm used in MAC protocol. Compare the back-off strategies as used in MACA and MACA-W protocols.

3. (a) Discuss the roles of the Care-of Address (CoA) in Mobile IP and the Binding Update process in MobileIPv6.

(b) Derive the relation between the cluster size and distance between co-channel cells.

(c) Assuming the interference is caused by 6 co-channel cells in a hexagonal cellular network and that all cells transmit with equal power, derive the minimum cluster size N if the Signal to Interference Ratio (SIR) is at least 18dB. 3+4+3

4. (a) In a scenario with high packet loss due to wireless interference, which variant of TCP (I-TCP, T-TCP, S-TCP, or M-TCP) would perform best? Justify your choice.

(b) Discuss the working principle of the variant of TCP used for the scenario in 4 (a).

(c) What are the advantages and disadvantages of the variant of TCP used for the scenario in 4 (a)? 2+5+3

5. (a) Discuss the role of power control in mitigating the near-far problem in GSM cellular systems.

(b) Show that the raw data rate of GSM Traffic channel TCH-F is 22.8 kbps.

(c) Discuss UMTS cellular architecture briefly. 3+3+4

6. (a) In DSDV, when topological changes make a route invalid, what will be the reaction of the protocol? Discuss through an example.

(b) "Choice of routing algorithm used in an ad-hoc network depends on the rate of topological change." Comment with justification.

(c) Describe the communication process in ZRP where the destination is not within the sender's zone. Illustrate through an example.

(d) State the problems that may affect the expected performance of ZRP. 2+2+4+2

7. (a) Can a loop (count to infinity problem) be formed while using AODV? Explain your answer through an example.

(b) How does AODV tackle the situation mentioned above?

(c) DSR caches the discovered routes for further data forwarding. Explain the strategy for refreshing the cache through an example. 3+3+4

8. Flooding is a simple strategy for distributing data to all sensor nodes in a network with limitations. Answer the following questions :

(a) State the limitations (Challenges) of flooding.

(b) Which of these can be addressed by gossiping, and how can they be addressed?

(c) How do sequence numbers contribute to reducing unnecessary transmissions? Are sequence numbers alone sufficient, and if not, what other information is needed to use them correctly?

(d) How does the SPIN family of protocols address the said challenges?

(e) Comment about addressing the challenges by Directed Diffusion compared to the strategies used by SPIN. 1+2+2+2+3